

Pratyuttar X

The final report — everything, in language you can explain

The final report

Everything about this project, in language you can explain

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This is the one document to read. Everything else we have written is either a shorter version of this or a form to fill in.

PART 1 · WHAT THE PROJECT IS

In one sentence

Pratyuttar X is a communication-safety project with two modules. One makes sure the right message gets out. The other makes sure the wrong message does not.

The two modules

Module 1 — FieldLink. An emergency alert that travels by radio when there is no mobile network. The base station checks the alert is genuine before trusting it, a person accepts responsibility for responding, and an acknowledgement goes back so the sender knows they were heard.

Module 2 — SATARK-AI. A laptop tool that checks a photo and caption *before* you post them, and shows what you are giving away without realising.

Why they belong together

They are the same problem from opposite ends.

	FieldLink	SATARK-AI
Cares about	A message that must get through	Information that should not get out
Prevents	A cry for help going unheard	A harmless post revealing too much
Works in	A physical emergency	Everyday digital life
Timing	During and after	Before
Location data	Must be precise	Must be discarded

PART 2 · THE PROBLEM

The story that explains everything

Tell this. Do not read it. It takes forty seconds.

A student posts three photos over three weeks.

Week one. A photo outside the school gate. Perfectly ordinary. But the photo *file* secretly carries the exact spot where it was taken — the phone wrote that in automatically, and the student has no idea.

Week two. A group photo. Blurry in the background is a board with the school name and a vehicle number.

Week three. A caption: *"Every Friday evening, same routine."*

Now add them up. Anyone who saw all three knows **where** this child is, **which** school, and **when** they will be there.

Nobody would look twice at any one of them. Together they are a schedule.

Why people cannot see this themselves

The evidence is invisible. Almost nobody knows a photo file carries their location. Almost nobody studies the background of their own photos.

It builds up slowly. You post one at a time, weeks apart. You never see them together.

Existing tools have the same blind spot. They each check one post. None shows the build-up.

The other half of the problem

Somebody needs urgent help where there is no signal. They press a button. The message may or may not get through — **and they have no way of knowing.**

That not-knowing is its own kind of harm, separate from the emergency itself. A basic panic button stops at "message sent". The real question is *"has anyone heard me?"*

Both halves in one sentence

When it matters most, the right information does not reach the right person — and the wrong information reaches everyone else.

PART 3 · MODULE 2 — SATARK-AI

What it does

Checks a photo and caption before you post. Runs on an ordinary Windows laptop. **No internet. Nothing sent anywhere.**

The simplest description: a spell-checker for privacy. It underlines the problem and lets you decide. It does not send, block or report.

The four checks

1 · What is hidden inside the photo file

Every photo from a phone carries invisible extra information — where it was taken, when, and on which device. We report whether it is there.

We never read the actual location. The number is thrown away inside the program before anything can show it. There is no way to make this software print where you were.

2 · What text can be read in the picture

Signboards. Gate names. Vehicle numbers. Notice boards. An ID card someone is holding. A Wi-Fi name or a QR code on a wall.

If the first look finds nothing, **it zooms in and looks again.** That second look is how it reads a small number plate in a busy photo.

3 · What the caption gives away

A trained model reads your words for place, routine, travel and group clues. Separate fixed rules catch the most dangerous shape of all:

a named place + a named person + a future time *"Today I will go to Riverside Mall with Sujata at 3pm."*

That tells a stranger where you will be, with whom, and when — **before it happens.** It also catches a vehicle number written into a caption.

4 · The pattern across several posts

This is the heart of it. It counts repetition — same place, same day, same hour.

One post at a place is a visit. Four posts at the same place, same day, same time, is **a schedule.**

What you see on the screen

One big coloured box — LOW, MEDIUM, HIGH or CRITICAL, with a plain sentence saying what that means.

What we found — the actual clues, in ordinary words.

What to do about it — specific suggestions tied to what was found.

Four buttons — Edit Before Sharing, Keep Private, Review Later, Mark as Demo Test.

You choose. The software never chooses.

Then the photo is deleted.

Three boxes, not one

The screen shows **what the AI decided, what the safety rules decided, and the final answer** — separately.

Sometimes they disagree. The AI says HIGH, the rules push it to CRITICAL, and you watch it happen.

"We do not hide the disagreement. An AI can be confidently wrong, and we would rather you saw that than trusted one number."

When it cannot check

If a photo has no readable text, no hidden data and no history, it does **not** say LOW. It says **"not enough to check."**

Because LOW on an empty scan looks like approval, when it only means we saw nothing.

This idea came from our own students during rehearsal. Say that.

PART 4 · MODULE 1 — FIELDLINK

What it does

Someone presses a button. The alert travels by **LoRa radio at 433 MHz** — no mobile network, no SIM, no tower.

The base station **checks the alert is genuine** before trusting it. A human operator accepts responsibility. **An acknowledgement returns**, so the sender knows help is coming.

Why it is not just a panic button

It verifies before it trusts. Every packet carries an authentication tag, checked before the alert becomes real. Invalid packets are shown as **rejections, with the reason** — not as alerts.

It measures the human. Every emergency project measures whether the message arrived. **We measure whether a person accepted responsibility, and how long they took.** Past 60 seconds, it escalates.

The measurements

Band	433 MHz — licence-free in India (WPC rules, 2022)
Spreading factor	SF12
Payload	67 bytes
Maximum range	up to 50 metres, line of sight
Verified at	10 metres (33 ft) — 7 to 8 seconds
Press to receipt	6 to 7 seconds (10 presses)
Computed radio time	4.27 seconds (2.79 alert + 1.48 acknowledgement)
Five rapid presses	33 seconds total — all five arrived
Weakest signal decoded	−119 dBm, SNR −17 dB

The three numbers you must be able to explain

"Why 33 seconds for five presses?"

*"33 divided by 5 is 6.6 seconds each — the same as a single press. So the alerts **queue rather than collide**, and nothing was lost. But it also means if five people press at once, the fifth waits half a minute. That is a real limitation of one channel at this spreading factor."*

"Why 6–7 seconds when the radio time is 4.27?"

"Two to three seconds are processing and GPS acquisition on top of the airtime. We have not yet separated them, and that is the next measurement we would take."

Never pretend the gap is not there. Naming it and saying what would resolve it is the strong answer.

"Why is 50 metres the same speed as 10 metres?"

"Distance does not affect timing at this scale. Radio covers 50 metres in less than a millionth of a second. The delay is airtime and processing, not travel."

PART 5 · DECIBELS AND SIGNAL-TO-NOISE

Most students cannot explain this. Learn it and you are ahead of the hall.

What a decibel is

A decibel is a ratio, not an amount. It says how many times bigger or smaller one thing is, on a squashed scale.

Change	Decibels
2× stronger	+3 dB
10× stronger	+10 dB
100×	+20 dB
1,000×	+30 dB
Half	−3 dB

So 30 dB means one thousand times. That is why losing 30 dB to a wrong antenna destroyed our range.

What dBm means

Decibels compared to one milliwatt.

- 0 dBm = 1 milliwatt
- **+20 dBm** = 100 milliwatts — our transmit power
- **−119 dBm** = an almost unimaginably tiny amount

What signal-to-noise ratio means

Every receiver hears background noise. **SNR is how far the signal stands above it.**

Positive — signal louder than noise. Normal. **Negative** — signal **buried underneath** the noise.

Ordinary radio cannot decode a negative SNR. We measured **−17 dB** — the signal was roughly **fifty times weaker than the noise** — and the message still came through.

How that is possible

LoRa does not send a steady tone. It sends a **chirp** — a signal sweeping across frequencies, like a slide whistle.

The receiver **knows the exact shape** of that sweep and slides its own copy along until they line up. Everything matching builds up; random noise does not.

The analogy to use: *"Imagine hearing one person in a noisy crowd. If they shout one word, you miss it. But if they slowly hum a tune you already know, you can pick it out even though it is quieter than the*

crowd — because you know what to listen for."

SF12 stretches each bit the longest, giving the most of this benefit — best range, deepest noise penetration. **The cost is speed**: 2.79 seconds for one 67-byte alert.

The link budget and our 30 dB

Add up everything gained and lost:

- Transmit power: **+20 dBm**
- Receiver sensitivity: about **−137 dBm**
- **Total budget: about 157 dB**

Loss over 20 metres at 433 MHz is about **51 dB**. So roughly 106 dB should have been spare — enough for hundreds of metres.

We were missing about 30 dB. No software fault loses a factor of a thousand. It had to be physical.

The cause: a **2.4 GHz WiFi antenna fitted to a 433 MHz radio**. At 433 MHz the element needs **17.3 cm**. A WiFi element is **3.1 cm** — about a fifth of what is needed. Most of the energy never left the antenna.

PART 6 · HOW A PHOTO CARRIES HIDDEN INFORMATION

A photo file is two things

The picture — and a block of information *about* the picture, written automatically without asking you.

That block is called **EXIF** (Exchangeable Image File Format).

What it can contain

Field	What it holds
GPS latitude and longitude	The exact spot the photo was taken
Date and time	The exact second the shutter fired
Make and model	The phone or camera used
Software	The editing app, if any
Camera settings	Exposure, ISO, focal length
Serial number	On some cameras, a unique device ID

Three things people get wrong

"Social media strips it." Most platforms do, on upload. **But the file on your phone still has it** — and that is the copy you send directly to someone by message or email.

"You would see it." You cannot see it in the picture. You need software that reads the file structure.

"It is only location." The **timestamp is often more revealing**. Three photos with timestamps establish a routine even with no location at all.

What we do with it

We report **whether** each field is present. **We never extract the coordinate** — it is thrown away inside the function that reads it.

"We report that location data is present. We never read what it says. Even if you asked, the software could not show you."

PART 7 · OCR — READING TEXT IN A PICTURE

Optical Character Recognition — finding text inside an image and turning it into letters a computer can work with. The same technology that reads a scanned document.

We use **Tesseract**, an open-source engine running entirely on the laptop.

How it works

1. **Clean the image** — grayscale, boost contrast, straighten
2. **Find the text** — locate regions that look like writing
3. **Read it** — match each shape against learned letterforms

Why Tesseract and not EasyOCR

EasyOCR downloads its models the first time it runs. On an exhibition table with no internet, that fails at the worst possible moment. Tesseract is one installer and works offline immediately.

Our two-pass reader

We tested a real cluttered photo — a student on a motorbike — and the standard single pass found **nothing at all**.

So the reader now **escalates**: if the first pass finds fewer than five words, it zooms in twice and switches to a mode for scattered text.

Result: 91.5% of evidence detected across 300 test images, with 5.6% false alarms on harmless ones.

PART 8 · WHERE THE AI IS

Two parts genuinely learned from examples. Everything else is ordinary code.

	Method	Job	Score
Caption reader	TF-IDF → Logistic Regression	What kind of clue your words contain	91.5%
Overall judge	Random Forest , 28 features	The final risk level	0.852 kappa

Not AI: reading hidden file data, reading picture text, the safety rules, counting repeats, the explanation. **And the final decision, which is a person.**

TF-IDF — turning words into numbers

A computer cannot work with letters, only numbers.

Term Frequency — how often a word appears here. **Inverse Document Frequency** — how rare it is everywhere else.

Multiply them: common words like *the* get almost no weight; distinctive ones get a lot.

We use word pairs too, so "every Friday" is its own feature — because the pair means something the two words separately do not.

Logistic Regression

Gives each word a **weight**, adds them up, converts the total into probabilities.

We chose it because you can read it — look at the weights and see exactly which words the model learned to care about.

Random Forest — the combination spotter

A decision tree asks yes/no questions: *Is GPS present? Is there a signboard? Does the caption mention a routine?* Each answer leads to the next.

One tree is unreliable — it latches onto accidents in the data.

A Random Forest builds hundreds of trees, each seeing a different slice of the data and a random subset of the questions. **Then they vote.**

Why it suits our problem: our risk comes from **combinations** — GPS *plus* a signboard *plus* a routine caption. Trees handle combinations naturally, because each branch stacks a condition on the one before. A method that just adds up scores would miss that.

Ours: 600 trees, 28 features, 4 levels.

Calibration — making the confidence honest

A model saying "80% confident" should be right about 80% of the time. **Raw scores usually are not.** We correct ours.

Result: calibration error 0.067 → 0.050.

Conformal prediction — knowing when to say nothing

When it cannot narrow to one level, it returns a **set** of possible levels with a guarantee: the true answer is inside at least 90% of the time.

Result: target 90%, achieved 91%.

Why kappa and not accuracy

Our four levels are **in order**. Saying HIGH when the truth is CRITICAL is a small mistake; saying LOW is a big one. **Plain accuracy treats them the same.**

So we report **quadratic weighted kappa**, which counts *how far* wrong.

0.852 kappa · 99.5% within one level · 1 in 200 off by more than one.

How we stopped the data being too easy

Our first caption model scored 100%. That is a warning, not a success — if a test is that easy, the score means nothing.

So we rebuilt it. Each story now gets **hidden truths** first, the answer comes from those, and the **visible clues are generated with noise** — a clue might show up in the file, or the picture, or only the caption, or nowhere.

Now the model has to work with incomplete evidence. That is the real task.

How we stopped it cheating

Data leakage is when test information reaches the model during training. It is the commonest mistake in student projects.

We split by **storyline**, not by row, so all versions of one story stay on the same side. **An assertion in the code fails the build** if that ever breaks.

PART 9 · THE PYTHON LIBRARIES

Library	What it does for us
scikit-learn	All machine learning — TF-IDF, logistic regression, random forest, calibration, every metric
Pillow (PIL)	Opening images, reading EXIF, cleaning images before OCR
pytesseract	The bridge to the Tesseract OCR engine
NumPy	Fast numerical arrays
pandas	Data tables — dataset, splits, results
Streamlit	The interface, in a browser on the laptop
joblib	Saving and loading trained models
matplotlib	Drawing the confusion matrix
piexif	Writing EXIF into our staged demo images
OpenCV	Webcam capture fallback only
pytest	Running our 17 tests

From Python itself: hashlib (integrity checks), secrets (random IDs), re (pattern matching), csv, json, pathlib, unicodedata

What we deliberately did not use

No PyTorch or TensorFlow. Deep learning needs far more data than our 1,000 examples, would overfit, and needs downloads that break our offline rule.

No cloud, no API, no language model. A tool built to prevent accidental exposure must not transmit your photo.

"Every library is free, open source, and runs on the laptop. Nothing calls out to the internet."

PART 10 · HOW THE CODE WAS WRITTEN

You will be asked. Answer straight.

"We designed the logic ourselves — what the system checks, which rules fire, how the levels are decided, and what it must refuse to do. We used AI assistance to write the code, which is normal practice in software work now. Then we tested it ourselves, and that testing found several real bugs which we fixed."

Why that answer is fine

Using AI assistance to write code is standard practice. Claiming otherwise would be the dishonest part, not the using.

What matters is whether you understand your system — and you can show that in ways nobody can fake:

- You can name every technique and say why it was chosen
- You can explain why kappa is right and accuracy is wrong
- You found the "am" matching inside "family" bug
- You worked out the 30 dB deficit and traced the physical cause
- You noticed $33 \div 5 = 6.6$ and understood what it means

None of that comes from a code generator.

***If pressed:** "The code was assisted. The design, the testing and the debugging were ours — and I can walk you through any part of this system and tell you why it works the way it does."*

PART 11 · WHAT WE CLAIM

The governing rule

Never say "we invented." Not once, about anything.

"We did not invent anything. We pointed an existing technique in the defensive direction, removed what could be misused, and measured our own work — including where it fails."

The four things that are genuinely ours

- 1. We turned an investigation tool into a self-protection tool.** Tools that work out somebody's routine from their posts already exist — built for an investigator, about someone else. We built the same analysis for the person themselves, before they post, and removed everything that could target anyone else.
- 2. We show the AI and the rules disagreeing.** Both answers on screen.
- 3. Our system can say "I could not check this."**
- 4. We measure whether a human took responsibility.** Standard in professional alarm companies; almost never in student projects.

What we did not invent

LoRa · HMAC-SHA256 · OCR · photo metadata · TF-IDF · random forests · calibration · conformal prediction · pattern-of-life analysis · response-time measurement

What we have not achieved

Encryption · protection against replayed messages · guaranteed delivery · testing on real posts · any connection between the two modules · certification · production readiness

Words never to use

invented · encrypted · fully secure · military-grade · unhackable · guaranteed · 100% · perfect · safe · certified · works anywhere · replaces emergency services

PART 12 · OUR WEAK POINTS — SAY THEM FIRST

1 · Our range is up to 50 metres

"LoRa can reach kilometres. Ours does not, and we know why. We calculated the link budget and found about 30 decibels missing — software cannot lose 30 dB, so it had to be physical. It was a 2.4 GHz WiFi antenna on a 433 MHz radio. The element needs 17.3 centimetres; a WiFi one is 3.1."

2 · Our AI has never been tested on real posts

"Everything it learned came from examples we wrote ourselves. So we know how well it learned our examples, and nothing about real ones. Testing against real posts, with permission, is our next step."

3 · The modules are not connected

"Separate hardware, separate code, no cable. That was a decision, not a gap. A radio packet carries 67 bytes and cannot carry a photo. Module 2's whole claim is that it has no network. And if you join two systems, a fault in one breaks the other."

PART 13 · MISTAKES WE FOUND OURSELVES

Volunteer these. They make everything else believable.

The AI that scored 100%. *"Our first caption model got everything right, and that worried us. If a test is that easy the score means nothing. So we rebuilt the whole dataset."*

The word "am". *"We tested our text reader on 300 images and found it flagging every harmless photo. The word 'am' was matching inside 'family' and 'programme'. Fixing it dropped false alarms from 26% to 6%."*

The photo that found nothing. *"We tried a real busy photo — a student on a motorbike — and our system found nothing at all. We rebuilt it to look twice. Now it reads the number plate."*

Our own students. *"They scanned a group photo, got LOW, and told us it was misleading. So we added a separate 'not enough to check' answer."*

The antenna. *"Our range was poor, so we did the maths and found 30 decibels missing."*

"We found all of those ourselves, before anyone else did, because we tested on messy things instead of only our own neat examples."

PART 14 · WHAT ALREADY EXISTS

Know these names. Agree first, then explain.

They say	You say
Meshtastic does encrypted LoRa mesh	"Yes, and better — longer range and encryption. But it sends messages. We measure whether a human accepted responsibility."
Maltego / Creepy aggregate someone's data	"Yes, and more powerfully. But built for an investigator looking at someone else. Ours is for the person, about themselves, before posting."
Phones already strip photo location	"Some do, on upload. The file on your phone still has it — and that is the copy you send directly."
Apple does on-device photo checking	"Yes, at a scale we could never reach. Theirs looks for one kind of content. Ours looks at how ordinary posts add up."
Alarm companies measure response time	"True, and standard in their industry. What we found is student emergency projects almost never do it."

PART 15 · SDGs, GROWTH AND DEPLOYMENT

Two primary goals

SDG 4 · Quality Education — we teach digital safety by showing it, not lecturing.

SDG 16 · Peace, Justice and Strong Institutions — one module makes sure a call for help is answered; the other reduces the exposure that makes a person easy to locate.

We also touch SDG 3, 9 and 11 — but would not claim those as primary. We do not list goals we cannot defend.

How it grows

Now — one laptop, one classroom, one alert unit. **Pilot** — school lab machines; supervised campus zones with trained responders. **Strengthen** — more languages; correct antenna; coverage tests. **Wider** — a phone version, where posting actually happens.

The honest limit

"Installing the software anywhere is easy. Before it could be used for real decisions rather than teaching, it needs testing against real posts labelled by real people. That is a research step, not a coding one."

PART 16 · THE SIX SENTENCES TO MEMORISE

1. "Pratyuttar X has two modules. One makes sure the right message gets out. The other makes sure the wrong message does not."
 2. "SATARK-AI is a spell-checker for privacy. It shows you what your photo is giving away before you post — and then you decide."
 3. "FieldLink sends an emergency alert by radio with no mobile network, checks it is genuine before trusting it, and tells the sender a person heard it."
 4. "Two parts learned from examples. Everything else is ordinary code, and I can tell you which is which for any part of the system."
 5. "We trained it on examples we wrote ourselves. So we know how well it learned our examples, and we do not know how it performs on real posts."
 6. "In an emergency, the danger is that nobody hears you. In everyday life, the danger is that too many people hear you. We built for both."
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PART 17 · TEST EACH OTHER

1. What does Pratyuttar X do, in one sentence?
 2. Tell the three-post story.
 3. What are the four things SATARK-AI checks?
 4. Where exactly is the AI?
 5. What is a decibel? What does 30 dB mean?
 6. What is SNR, and why is -17 dB remarkable?
 7. Why did five presses take 33 seconds?
 8. Why 6–7 seconds when the radio time is 4.27?
 9. What is EXIF? Name four things it contains.
 10. What is OCR, and why Tesseract?
 11. What is a Random Forest, and why did we choose it?
 12. Why kappa instead of accuracy?
 13. What is data leakage, and how did we prevent it?
 14. Name five Python libraries and what each does.
 15. Did you write the code yourselves?
 16. Are the two modules connected? Why not?
 17. Why is your range only 50 metres?
 18. Has the AI been tested on real posts?
 19. Name three mistakes you found yourselves.
 20. What is the closing line?
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PART 18 · THE LAST THING

You do not need to sound impressive. **You need to sound like two people who built something, tested it properly, and know exactly where it breaks.**

A judge has heard many teams say their project is amazing. Very few have heard a team say *"here is where ours fails, and here is how we found out."*

That is your advantage. Use it.

And when you do not know: *"I do not know — that is a good question and I would need to check."* That answer never loses marks. A confident guess always does.